

Phase 2 Modeling (Hima Varsha)

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```
#####-----DATA PREPARATION-----  
#Load the dataset  
data <- read_excel("merged_counties_median_income.xlsx")  
head(data)  
str(data)  
summary(data)  
  
#Rename Columns to make modeling easier  
data <- data%>%  
  rename(  
    Geographic_Area = `Geographic Area`,  
    State_FIPS = `State FIPS Code`,  
    County_FIPS = `County FIPS Code`,  
    Unemployment_Rate = `Unemployment Rate (%)`,  
    Labor_Force = `Labor Force`  
  )  
  
names(data)  
  
#Check missing values  
colSums(is.na(data))  
  
#New Added step : Feature engineering  
# Step : Feature Engineering (Added for deeper insights)  
data <- data %>%  
  mutate(  
    Employment_Rate = Employed / Labor_Force,  
    Unemployment_Share = Unemployed / Labor_Force  
  )
```

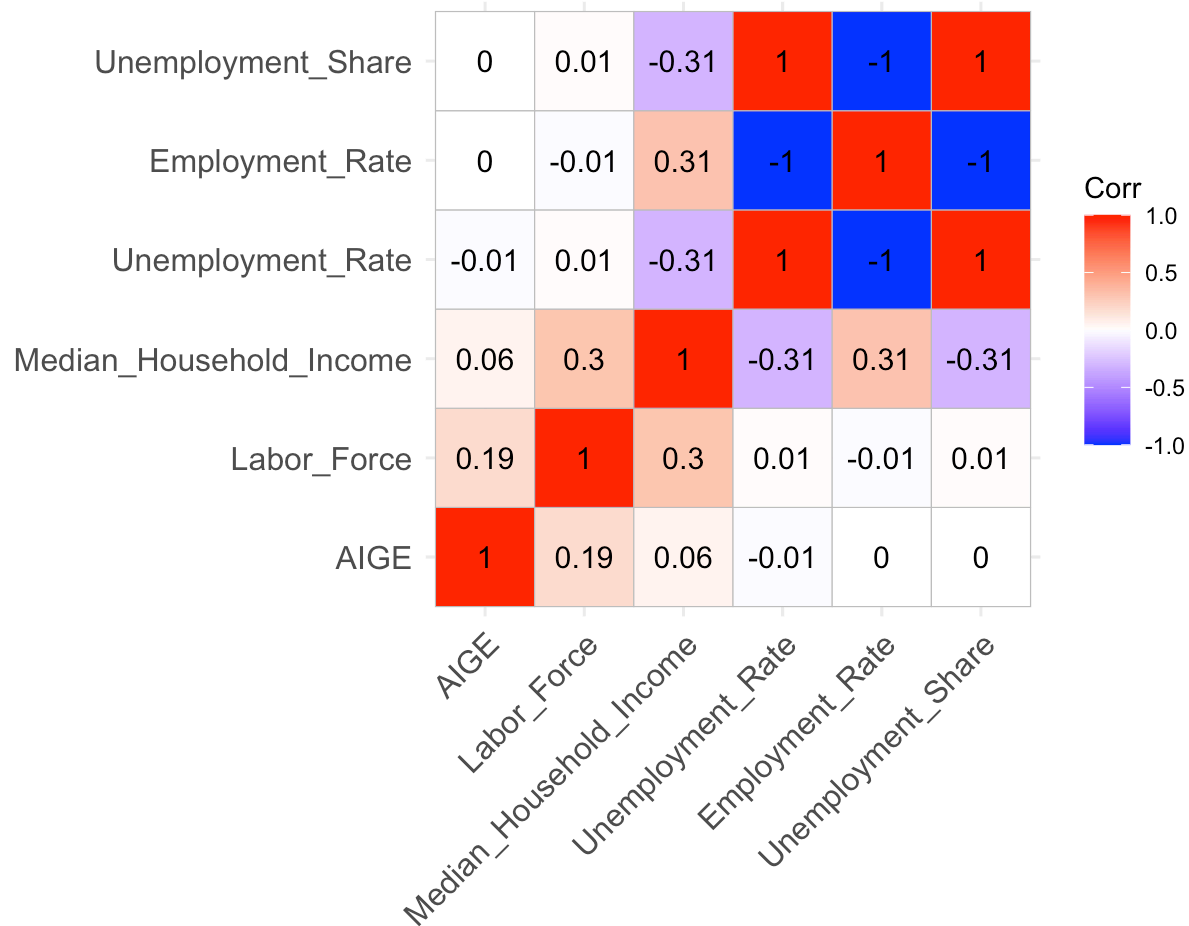


```

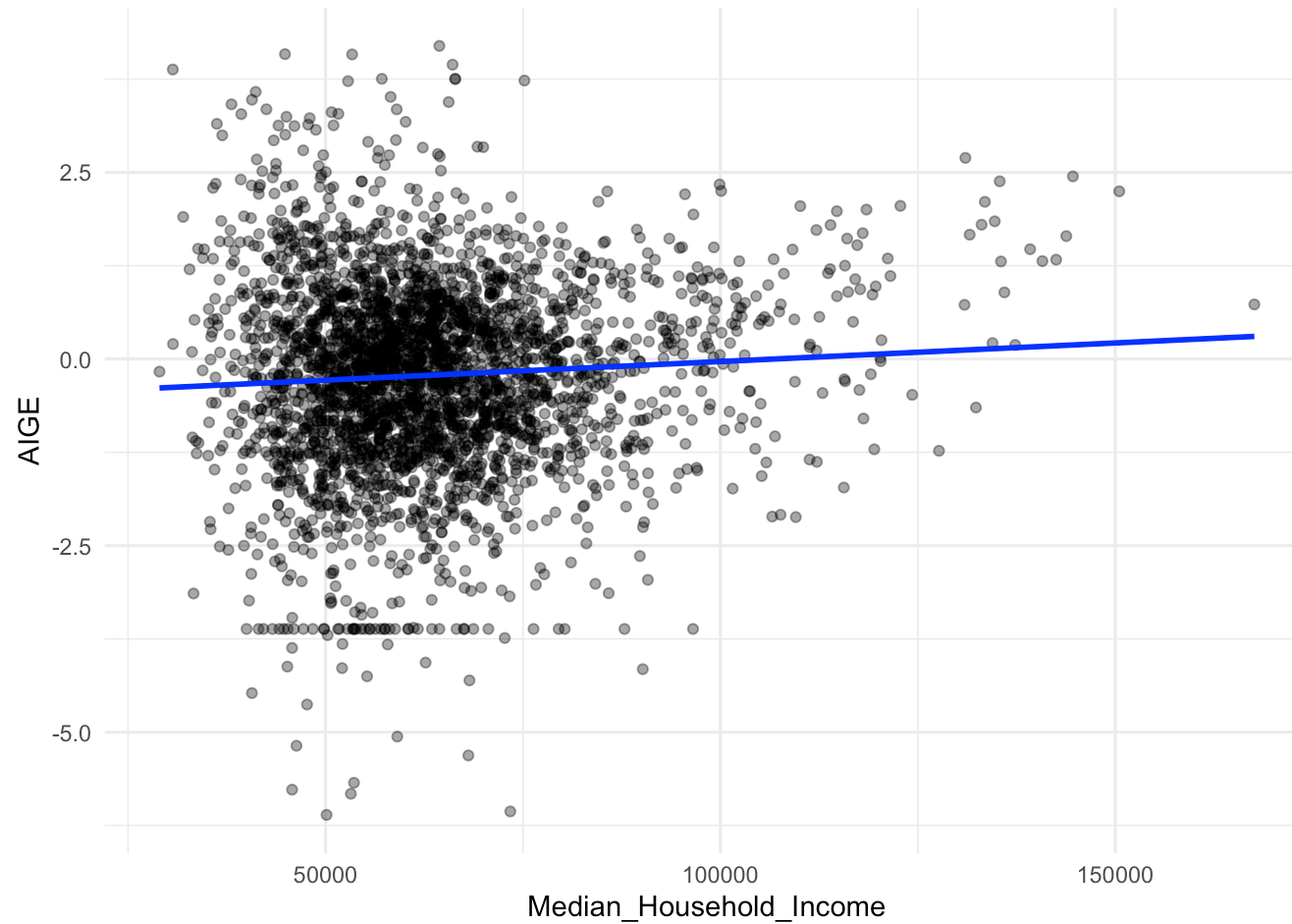
##      Employed      Unemployed      Unemployment Rate (%)
## Min.   :    92   Min.   :    3   Min.   : 1.300
## 1st Qu.:   4500   1st Qu.:   184   1st Qu.: 3.200
## Median :  11158   Median :   461   Median : 3.800
## Mean   :  51410   Mean   :  2161   Mean   : 3.977
## 3rd Qu.:  31101   3rd Qu.:  1255   3rd Qu.: 4.500
## Max.   :4812580   Max.   :297174   Max.   :18.400
## Median_Household_Income
## Min.   : 28972
## 1st Qu.: 52457
## Median : 60780
## Mean   : 63206
## 3rd Qu.: 70495
## Max.   :167605
## [1] "Geographic_Area"      "AIGE"
## [3] "FIPS"                    "State_FIPS"
## [5] "County_FIPS"             "State"
## [7] "Year"                    "Labor_Force"
## [9] "Employed"                "Unemployed"
## [11] "Unemployment_Rate"       "Median_Household_Income"
##      Geographic_Area      AIGE      FIPS
##      0      0      0
##      State_FIPS      County_FIPS      State
##      0      0      0
##      Year      Labor_Force      Employed
##      0      0      0
##      Unemployed      Unemployment_Rate      Median_Household_Income
##      0      0      0

```

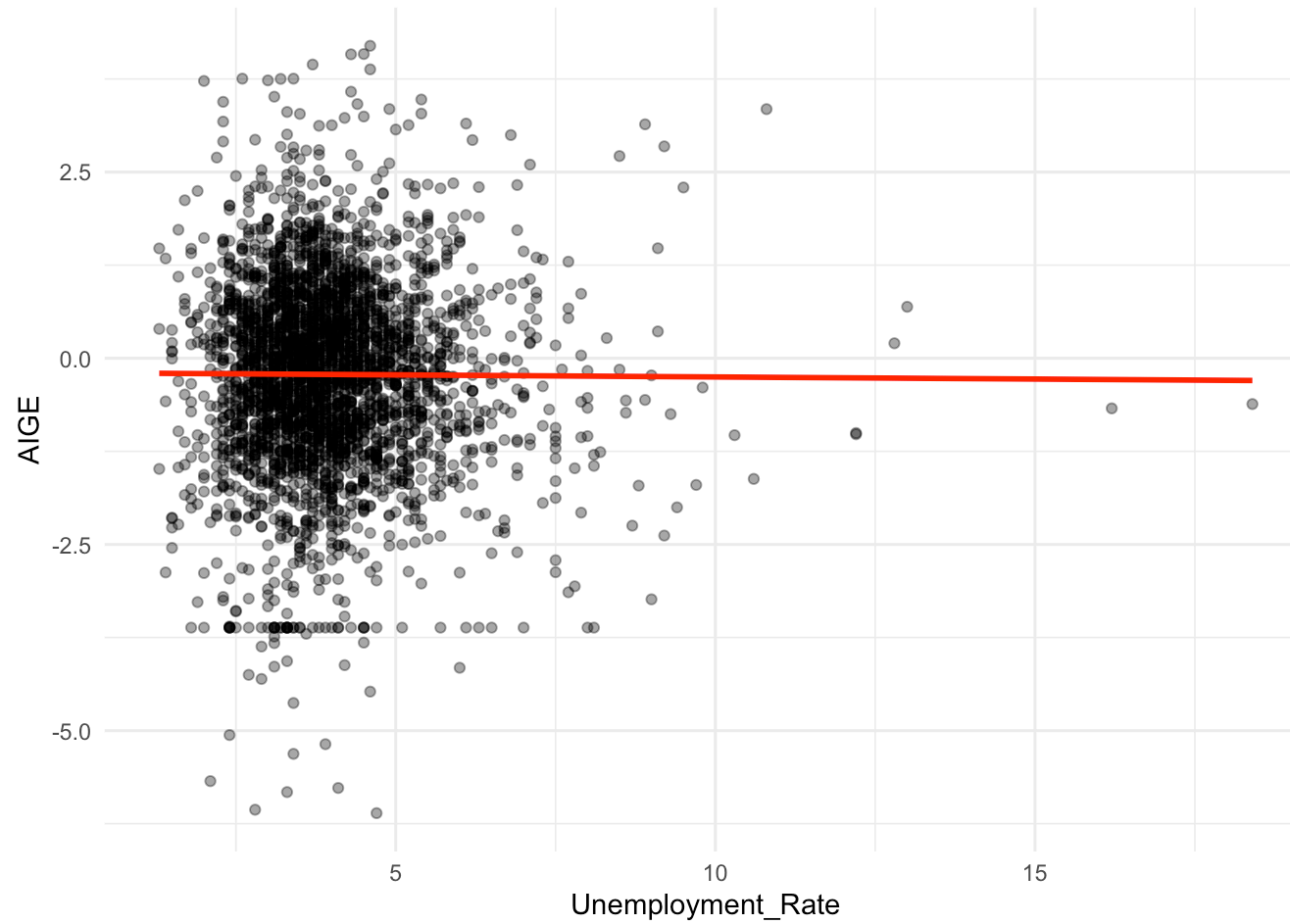
```
#####-----Exploratory Data Analysis (EDA)-----  
# Updated numeric variable set (clean)  
numeric_vars <- data %>%  
  select(  
    AIGE,  
    Labor_Force,  
    Median_Household_Income,  
    Unemployment_Rate,  
    Employment_Rate,  
    Unemployment_Share  
  )  
  
# Correlation matrix  
cor_matrix <- cor(numeric_vars)  
cor_matrix  
  
#Correlation Heatmap  
ggcorrplot(cor_matrix, lab = TRUE)
```



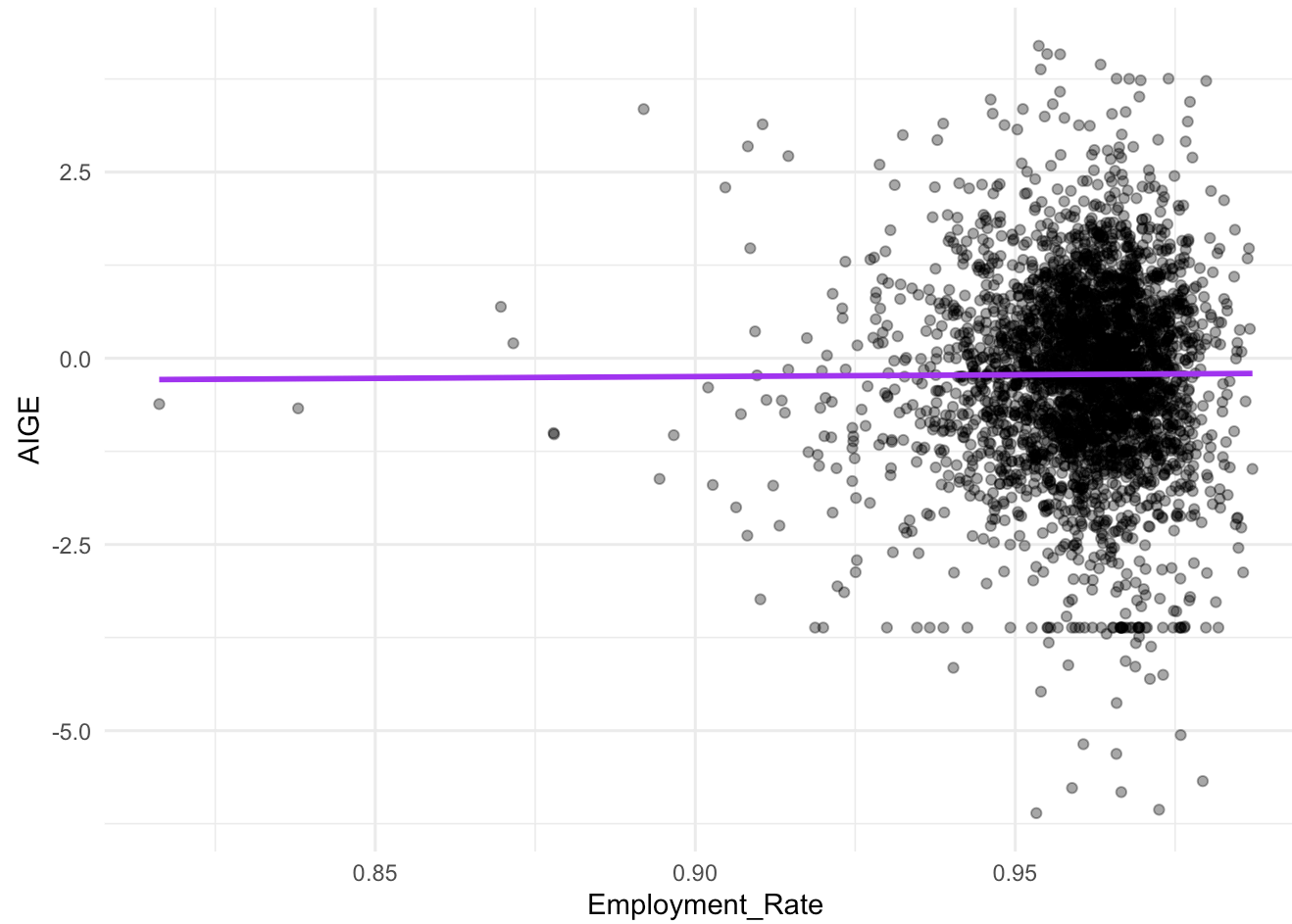
```
#Scatter Plots
#For AIGE vs Income
ggplot(data, aes(x = Median_Household_Income, y = AIGE)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se= FALSE, color = "blue") +
  theme_minimal()
```



```
#For AIGE vs Unemployment Rate  
ggplot(data, aes(x = Unemployment_Rate, y = AIGE)) +  
  geom_point(alpha = 0.4) +  
  geom_smooth(method = "lm", se = FALSE, color = "red") +  
  theme_minimal()
```



```
#For AIGE vs Employment Rate
ggplot(data, aes(x = Employment_Rate, y = AIGE)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "purple") +
  theme_minimal()
```



```
#For AIGE vs Unemployment Share  
ggplot(data, aes(x = Unemployment_Share, y = AIGE)) +  
  geom_point(alpha = 0.4) +  
  geom_smooth(method = "lm", se = FALSE, color = "darkgreen") +  
  theme_minimal()
```




##	AIGE	Labor_Force	Median_Household_Income
## AIGE	1.000000000	0.18982660	0.06340608
## Labor_Force	0.189826604	1.000000000	0.30206159
## Median_Household_Income	0.063406078	0.30206159	1.000000000
## Unemployment_Rate	-0.005636625	0.01300777	-0.31115137
## Employment_Rate	0.004780838	-0.01332141	0.31142903
## Unemployment_Share	-0.004780838	0.01332141	-0.31142903
##	Unemployment_Rate	Employment_Rate	Unemployment_Share
## AIGE	-0.005636625	0.004780838	-0.004780838
## Labor_Force	0.013007768	-0.013321413	0.013321413
## Median_Household_Income	-0.311151371	0.311429031	-0.311429031
## Unemployment_Rate	1.000000000	-0.999747827	0.999747827
## Employment_Rate	-0.999747827	1.000000000	-1.000000000
## Unemployment_Share	0.999747827	-1.000000000	1.000000000

```

#####----- REGRESSION MODEL-----
# Selecting modeling variables
model_data <- data %>%
  select(AIGE, Median_Household_Income, Unemployment_Rate, Labor_Force)

# Fit the Linear regression model
reg_model <- lm(AIGE ~ Median_Household_Income + Unemployment_Rate + Labor_Force,
  data = model_data)
summary(reg_model)

# Model selection (Backward, Forward, Stepwise)
#Backward Elimination
backward_model <- step(reg_model, direction = "backward", trace = TRUE)

#Forward Selection
null_model <- lm(AIGE ~ 1, data = model_data)
forward_model <- step(
  null_model,
  direction = "forward",
  scope = ~ Median_Household_Income + Unemployment_Rate + Labor_Force,
  trace = TRUE
)

#Stepwise (Both Directions)
stepwise_model <- step(
  null_model,
  scope = ~Median_Household_Income + Unemployment_Rate + Labor_Force,
  trace = TRUE
)

# AIC Comparison Table
aic_table <- data.frame(
  Model = c("Full Model", "Backward", "Forward", "Stepwise"),
  AIC = c(
    AIC(reg_model),

```

```
    AIC(backward_model),  
    AIC(forward_model),  
    AIC(stepwise_model)  
  )  
)  
  
print(aic_table)
```

```
##
## Call:
## lm(formula = AIGE ~ Median_Household_Income + Unemployment_Rate +
##     Labor_Force, data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.8092 -0.7197  0.0610  0.7536  4.4883
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.854e-01  1.398e-01  -2.041   0.0413 *
## Median_Household_Income  3.409e-07  1.533e-06   0.222   0.8240
## Unemployment_Rate    -6.729e-03  1.857e-02  -0.362   0.7171
## Labor_Force        1.368e-06  1.345e-07  10.171  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.254 on 3128 degrees of freedom
## Multiple R-squared:  0.03612,    Adjusted R-squared:  0.03519
## F-statistic: 39.07 on 3 and 3128 DF,  p-value: < 2.2e-16
##
## Start:  AIC=1423.72
## AIGE ~ Median_Household_Income + Unemployment_Rate + Labor_Force
##
##              Df Sum of Sq    RSS    AIC
## - Median_Household_Income  1      0.078 4921.9 1421.8
## - Unemployment_Rate        1      0.207 4922.1 1421.8
## <none>                      4921.9 1423.7
## - Labor_Force              1    162.762 5084.6 1523.6
##
## Step:  AIC=1421.77
## AIGE ~ Unemployment_Rate + Labor_Force
##
##              Df Sum of Sq    RSS    AIC
## - Unemployment_Rate  1      0.336 4922.3 1420.0
## <none>                4921.9 1421.8
## - Labor_Force        1    184.174 5106.1 1534.8
##
```

```

## Step: AIC=1419.98
## AIC ~ Labor_Force
##
##           Df Sum of Sq    RSS    AIC
## <none>                4922.3 1420.0
## - Labor_Force  1      184 5106.3 1532.9
## Start: AIC=1532.92
## AIC ~ 1
##
##           Df Sum of Sq    RSS    AIC
## + Labor_Force      1  184.001 4922.3 1420.0
## + Median_Household_Income 1   20.529 5085.8 1522.3
## <none>                5106.3 1532.9
## + Unemployment_Rate  1    0.162 5106.1 1534.8
##
## Step: AIC=1419.98
## AIC ~ Labor_Force
##
##           Df Sum of Sq    RSS    AIC
## <none>                4922.3 1420.0
## + Unemployment_Rate  1   0.33556 4921.9 1421.8
## + Median_Household_Income 1   0.20681 4922.1 1421.8
## Start: AIC=1532.92
## AIC ~ 1
##
##           Df Sum of Sq    RSS    AIC
## + Labor_Force      1  184.001 4922.3 1420.0
## + Median_Household_Income 1   20.529 5085.8 1522.3
## <none>                5106.3 1532.9
## + Unemployment_Rate  1    0.162 5106.1 1534.8
##
## Step: AIC=1419.98
## AIC ~ Labor_Force
##
##           Df Sum of Sq    RSS    AIC
## <none>                4922.3 1420.0
## + Unemployment_Rate  1    0.336 4921.9 1421.8
## + Median_Household_Income 1    0.207 4922.1 1421.8
## - Labor_Force      1  184.001 5106.3 1532.9

```

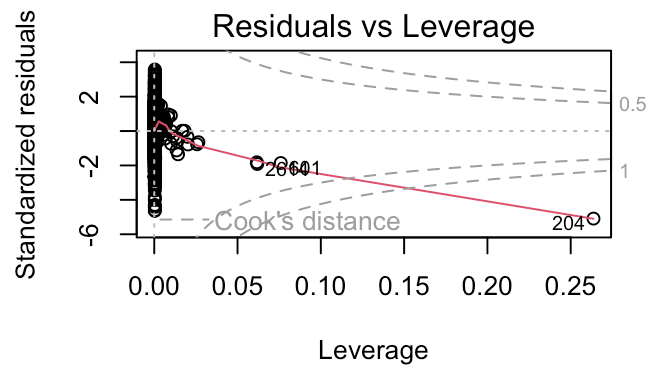
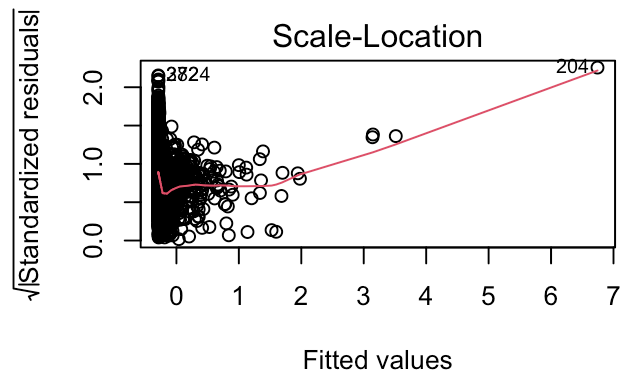
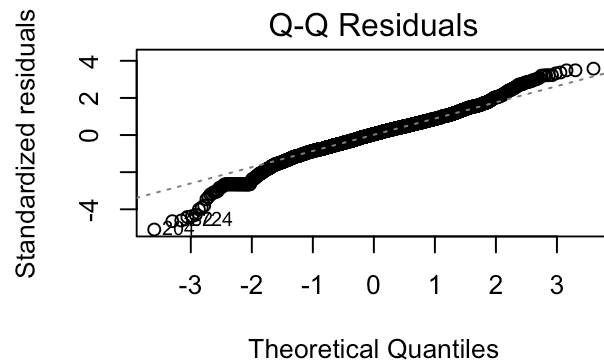
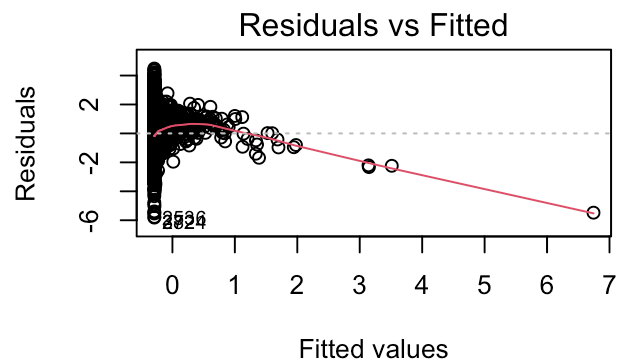
```
##          Model      AIC
## 1 Full Model 10313.95
## 2   Backward 10310.21
## 3    Forward 10310.21
## 4   Stepwise 10310.21
```

```
# Train / Test Split + RMSE
set.seed(123)

train_index <- createDataPartition(model_data$AIGE, p= 0.8, list = FALSE)
train_data <- model_data[train_index, ]
test_data <- model_data[-train_index, ]

#Train the final regression model on training data : Using the selected variable labor_force
final_reg <- lm(AIGE ~ Labor_Force, data = train_data)
summary(final_reg)

#Regression diagnostic plots
par(mfrow = c(2,2))
plot(backward_model)
```




```
par(mfrow = c(1, 1))

#Predict
pred <- predict(final_reg, newdata = test_data)
actual <- test_data$AIGE

#Compute RMSE, Rsquared, MAE
core_metrics <- postResample(pred = pred, obs = actual)
core_metrics

#Extra Metrics (ME & MAPE)
ME <- mean(actual - pred)
MAPE <- mean(abs((actual - pred) / actual)) * 100

performance <- data.frame(
  RMSE = core_metrics["RMSE"],
  R2 = core_metrics["Rsquared"],
  MAE = core_metrics["MAE"],
  ME = ME,
  MAPE = MAPE
)
performance
```

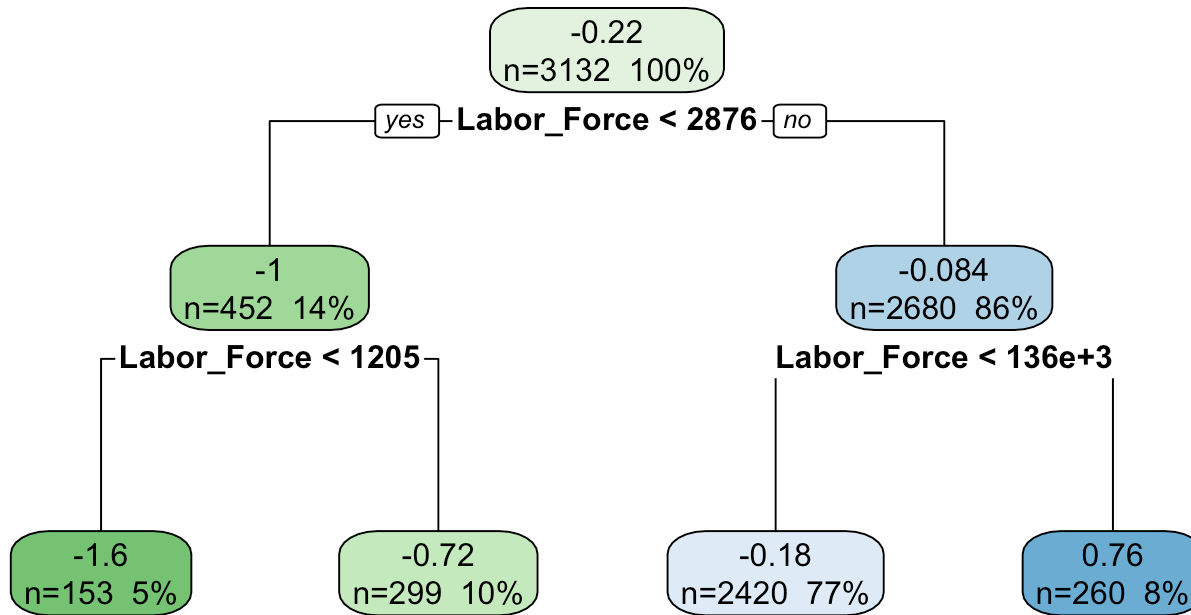
```
##
## Call:
## lm(formula = AIGE ~ Labor_Force, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.8209 -0.7200  0.0639  0.7615  4.4822
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.882e-01  2.592e-02  -11.12   <2e-16 ***
## Labor_Force  1.240e-06  1.335e-07    9.29   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.246 on 2506 degrees of freedom
## Multiple R-squared:  0.03329,    Adjusted R-squared:  0.0329
## F-statistic: 86.3 on 1 and 2506 DF,  p-value: < 2.2e-16
##
##      RMSE   Rsquared      MAE
## 1.2885519 0.06091298 0.96381671
##      RMSE      R2      MAE      ME MAPE
## RMSE 1.288555 0.06091298 0.9638167 0.02211412 Inf
```

```
#####----- DECISION TREE MODEL-----
#Fitting the Tree Model
tree_data <- model_data
tree_model <- rpart(
  AIGE ~ Median_Household_Income + Unemployment_Rate + Labor_Force,
  data = tree_data,
  method = "anova"
)

tree_model

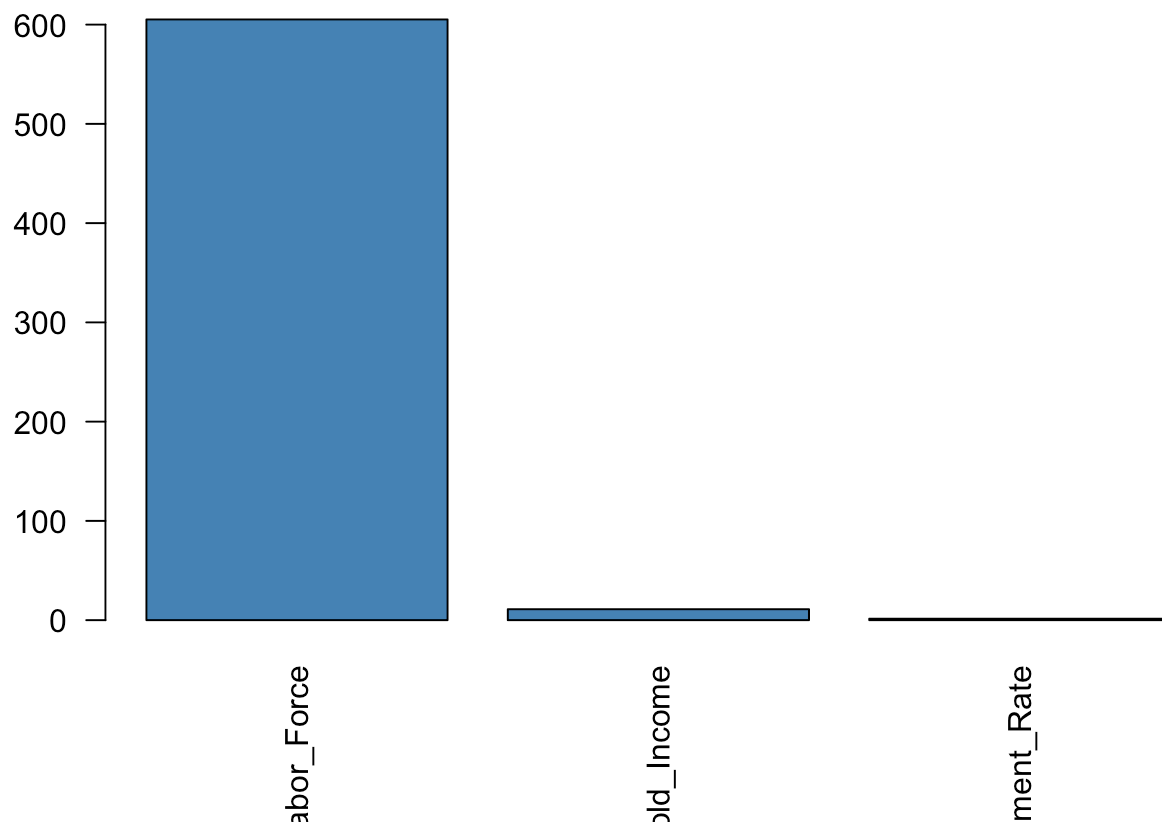
#Plotting the tree
rpart.plot(tree_model, type = 2, extra = 101, fallen.leaves = TRUE,
  box.palette = "GnBu", main = "Decision Tree for AIGE")
```

Decision Tree for AIGE



```
#Variable importance
vip <- tree_model$variable.importance
barplot(vip, main = "Variable Importance (Decision Tree)",
        col = "steelblue", las = 2)
```

Variable Importance (Decision Tree)



```

#Predict on test data
tree_pred <- predict(tree_model, newdata = test_data)

#Compute performance metrics
tree_core <- postResample(pred = tree_pred, obs = actual)
tree_core

#Extra metrics
tree_ME <- mean(actual - tree_pred)
tree_MAPE <- mean(abs((actual - tree_pred) / actual)) * 100

tree_performance <- data.frame(
  RMSE = tree_core["RMSE"],
  R2 = tree_core["Rsquared"],
  MAE = tree_core["MAE"],
  ME = tree_ME,
  MAPE = tree_MAPE
)

tree_performance

```

```

## n= 3132
##
## node), split, n, deviance, yval
##      * denotes terminal node
##
## 1) root 3132 5106.2840 -0.21737230
##   2) Labor_Force< 2876 452 1878.1180 -1.00539400
##     4) Labor_Force< 1205 153 898.9273 -1.55952900 *
##     5) Labor_Force>=1205 299 908.1691 -0.72183950 *
##   3) Labor_Force>=2876 2680 2900.1450 -0.08446716
##     6) Labor_Force< 135915 2420 2572.5050 -0.17538100 *
##     7) Labor_Force>=135915 260 121.4646 0.76173080 *
##      RMSE   Rsquared   MAE
## 1.26812070 0.07910608 0.93091081
##      RMSE      R2      MAE      ME MAPE
## RMSE 1.268121 0.07910608 0.9309108 0.02603303 Inf

```

```
#####----- NEURAL NETWORK MODEL-----
```

```
#Normalize(scale) the variables
```

```
normalize <- function(x) {  
  return((x - min(x)) / (max(x) - min(x)))  
}
```

```
nn_data <- model_data %>%
```

```
  mutate(  
    Median_Household_Income = normalize(Median_Household_Income),  
    Unemployment_Rate = normalize(Unemployment_Rate),  
    Labor_Force = normalize(Labor_Force)  
  )
```

```
# Train the neural network
```

```
nn_train <- nn_data[train_index, ]  
nn_test <- nn_data[-train_index, ]
```

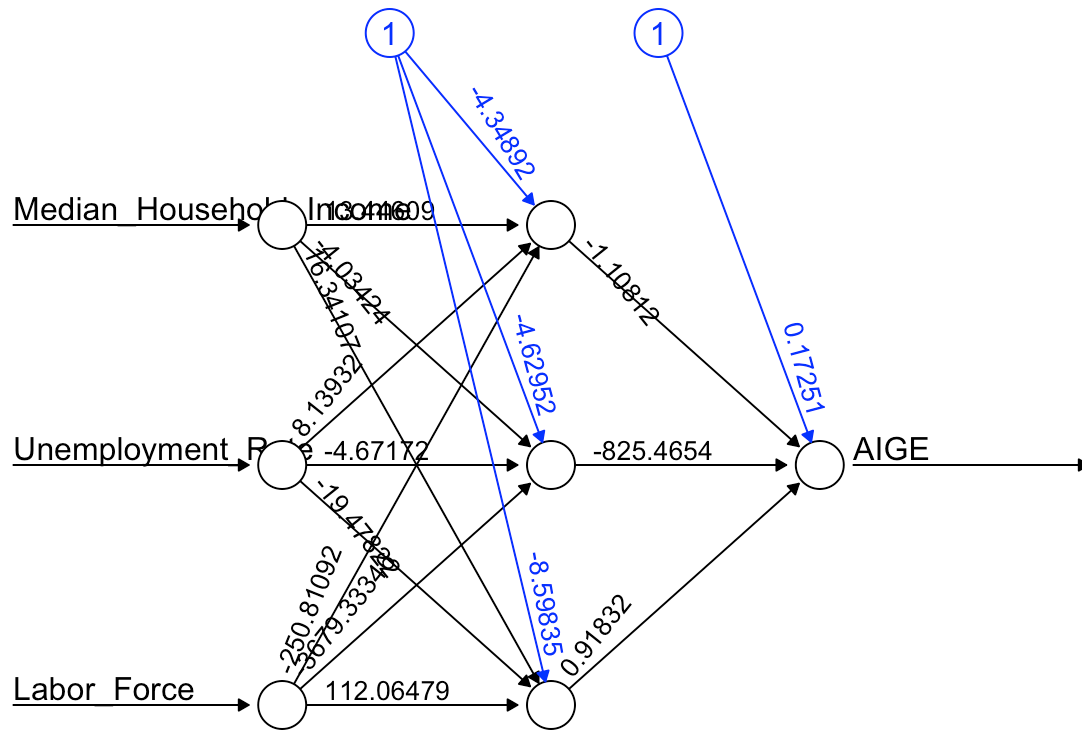
```
#Neural network with 1 hidden layer (3 neurons)
```

```
set.seed(123)
```

```
nn_model_1 <- neuralnet(  
  AIGE ~ Median_Household_Income + Unemployment_Rate + Labor_Force,  
  data = nn_train,  
  hidden = 3,  
  linear.output = TRUE  
)
```

```
#Plot the neural network
```

```
plot(nn_model_1, rep = "best")
```



Error: 1699.699983 Steps: 36785

```

# Predict on test data (Neural Net with 1 hidden layer)
nn_pred_raw <- compute(
  nn_model_1,
  nn_test[, c("Median_Household_Income", "Unemployment_Rate", "Labor_Force")]
)

nn_pred <- as.vector(nn_pred_raw$net.result)
actual_nn <- nn_test$AIGE # AIGE was not normalized

# Core metrics (RMSE, R2, MAE)
nn_core <- postResample(pred = nn_pred, obs = actual_nn)
nn_core

# Extra metrics (ME, MAPE)
nn_ME <- mean(actual_nn - nn_pred)
nn_MAPE <- mean(abs((actual_nn - nn_pred) / actual_nn)) * 100

nn_performance <- data.frame(
  RMSE = nn_core["RMSE"],
  R2    = nn_core["Rsquared"],
  MAE   = nn_core["MAE"],
  ME    = nn_ME,
  MAPE  = nn_MAPE
)

nn_performance

```

```

##          RMSE   Rsquared         MAE
## 1.25702074 0.09798105 0.89730443
##          RMSE          R2         MAE          ME MAPE
## RMSE 1.257021 0.09798105 0.8973044 0.03529155 Inf

```

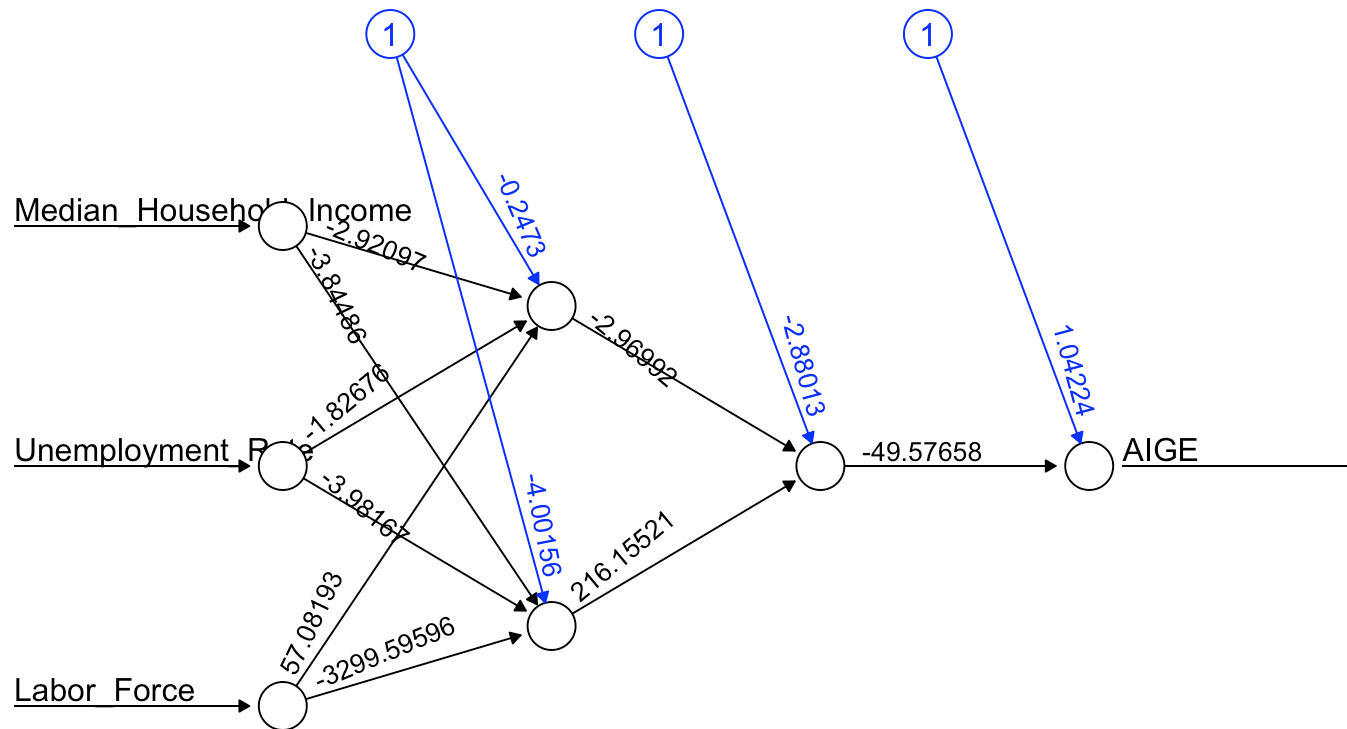


```

# Neural Network with 2 Hidden Layers
set.seed(123)
nn_model2 <- neuralnet(
  AIGE ~ Median_Household_Income + Unemployment_Rate + Labor_Force,
  data = nn_train,
  hidden = c(2,1),          # lighter 2-layer network
  linear.output = TRUE,
  stepmax = 2e5
)

#Plot
plot(nn_model2, rep="best")

```



Error: 1720.956998 Steps: 38476

```

#Metrics
nn_pred_raw2 <- compute(
  nn_model2,
  nn_test[, c("Median_Household_Income", "Unemployment_Rate", "Labor_Force")]
)
nn_pred2 <- as.vector(nn_pred_raw2$net.result)

nn_core2 <- postResample(pred = nn_pred2, obs = nn_test$AIGE)
nn_core2

nn_ME2 <- mean(nn_test$AIGE - nn_pred2)
nn_MAPE2 <- mean(abs((nn_test$AIGE - nn_pred2) / nn_test$AIGE)) * 100

nn_performance2 <- data.frame(
  RMSE = nn_core2["RMSE"],
  R2 = nn_core2["Rsquared"],
  MAE = nn_core2["MAE"],
  ME = nn_ME2,
  MAPE = nn_MAPE2
)
nn_performance2

```

```

##          RMSE  Rsquared          MAE
## 1.26384796 0.08810434 0.91124606
##          RMSE          R2          MAE          ME MAPE
## RMSE 1.263848 0.08810434 0.9112461 0.02930318 Inf

```

```

model_compare <- data.frame(
  Model = c("Regression", "Decision Tree", "Neural Net (1 layer)", "Neural Net (2 layers)"),
  RMSE = c(1.2886, 1.2681, 1.2570, 1.2638),
  MAE = c(0.9638, 0.9309, 0.8973, 0.9112),
  R2 = c(0.0609, 0.0791, 0.0980, 0.0881),
  ME = c(0.0221, 0.0260, 0.0353, 0.0293)
)

model_compare
write.csv(model_compare, "model_compare.csv", row.names=FALSE)

```

##	Model	RMSE	MAE	R2	ME
## 1	Regression	1.2886	0.9638	0.0609	0.0221
## 2	Decision Tree	1.2681	0.9309	0.0791	0.0260
## 3	Neural Net (1 layer)	1.2570	0.8973	0.0980	0.0353
## 4	Neural Net (2 layers)	1.2638	0.9112	0.0881	0.0293